BASE CHANGE FOR DIRECT IMAGES ON A MORPHISM OF SIMPLICIAL SITES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs lemma-restriction-injective-to-component-site, lemma-morphism-simplicial-sites.

Editorial note. The label lemma-direct-image-morphism-simplicial-sites names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- K-injective restriction. Same unit, Lemma lemma-restriction-injective-to-component-site. g_n^{-1} of a K-injective complex is K-injective on the component.
- Levelwise direct image. Same unit, Lemma lemma-morphism-simplicial-sites. The levelwise identity (g'_n)^{-1}h_{total,*} = h_{n,*}g_n^{-1} on sheaves underlying the derived statement.

Source reference. Original reference 0D96: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1273–1273. Prerequisite pointer invoked at proof line(s) 1354; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WH: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1086–1086. Prerequisite pointer invoked at proof line(s) 1364; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With notation and hypotheses as in Lemma 0D96. For $K \in D(\mathcal{C}_{total})$ we have $(g'_n)^{-1}Rh_{total,*}K = Rh_{n,*}g_n^{-1}K$.*

Proof. Let $\mathcal{I}^\bullet$ be a K-injective complex on $\mathcal{C}_{total}$ representing K. Then $g_n^{-1}K$ is represented by $g_n^{-1}\mathcal{I}^\bullet = \mathcal{I}_n^\bullet$ which is K-injective by Lemma 09WH. We have $(g'_n)^{-1}h_{total,*}\mathcal{I}^\bullet = h_{n,*}g_n^{-1}\mathcal{I}_n^\bullet$ by Lemma 0D96 which gives the desired equality. □

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